Developmental interactions between oscillatory and spontaneous neural activity support improvements in inhibitory control into adulthood

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BACKGROUND

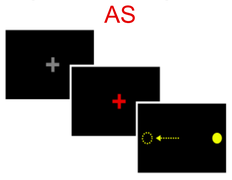
- Adolescence is a critical period for the development of cognitive control occurring in parallel with maturation of executive brain systems[1,2].
- A key component of cognitive control is inhibitory control- the ability to generate task-compatible responses while suppressing reflexive and incompatible responses[2].
- Oscillatory neural activity when preparing to inhibit a response is crucial to successful response inhibition in both humans[3,4] and non-human primates[5]
- Neural activity can be characterized as evoked (stimulus locked) or spontaneous (non-stimulus locked) activity. Spontaneous activity has been found to decrease with age[9,10], supporting higher fidelity neural signaling.

This project aims to test the hypothesis that there are age-related decreases in spontaneous activity during task that support enhanced transmission of task-relevant oscillatory activity in frontal cortex, contributing to developmental improvements in inhibitory control.

METHODS

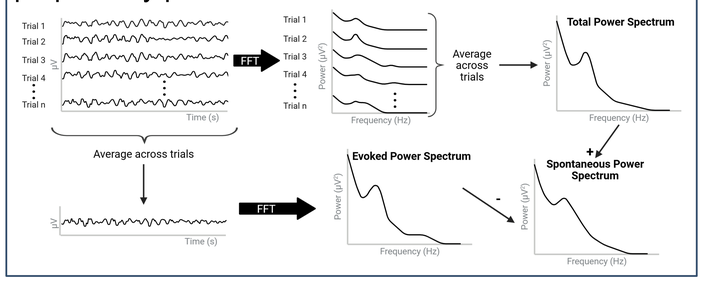
Participants: 144 (71F) 10-34 yos completed EEG sessions in a longitudinal study with up to 3 timepoints (256 total EEG sessions)

Oculomotor Tasks: Each participant completed 40 trials of both the anti-saccade (AS) task and visually-guided saccade (VGS) task.



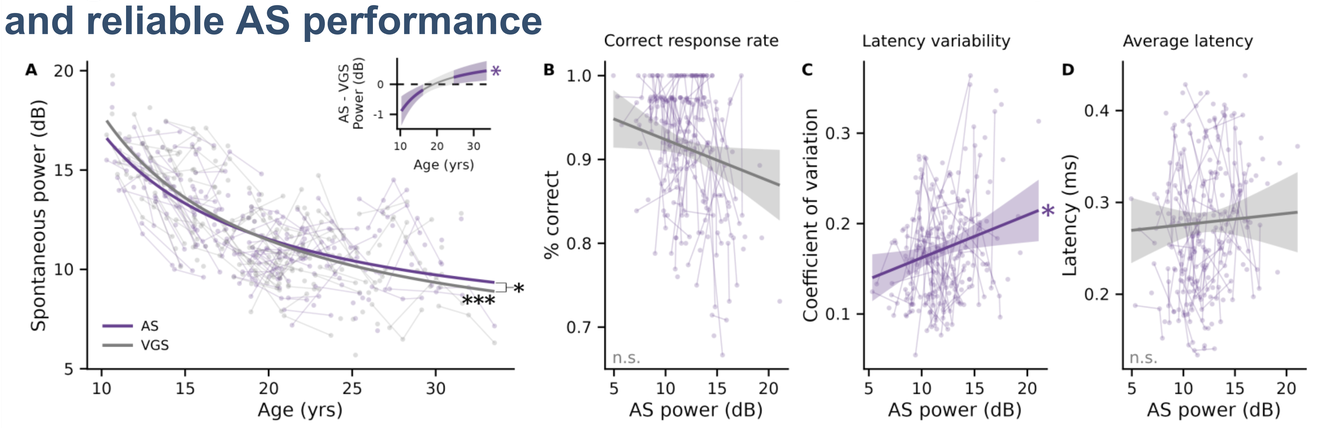
EEG: Data were collected during AS and VGS tasks. We computed event-related spectral perturbation (ERSP) using a baseline of 700 to 400 ms prior to onset of the preparatory period across frontal electrodes[3,4]. We used threshold-free cluster enhancement permutation analysis to identify significant time-frequency clusters.

Spontaneous Activity: Following prior approaches[9], spontaneous power was computed based on activity that was not phase-locked to the onset of the preparatory period.



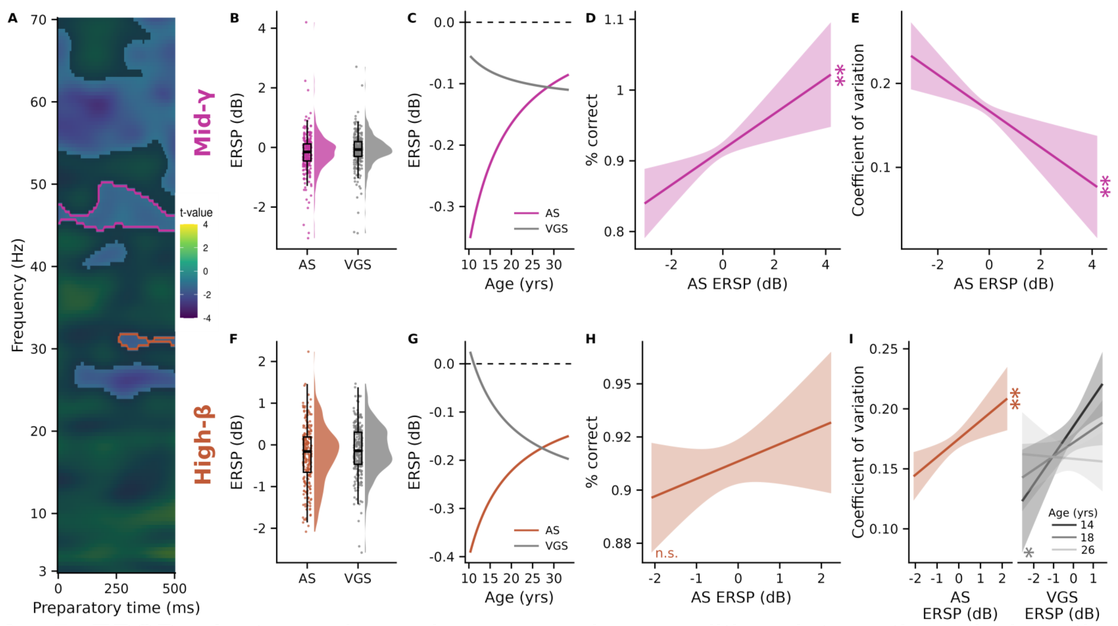
RESULTS

1. Age-related decreases in broadband spontaneous power are associated with more accurate and reliable AS performance



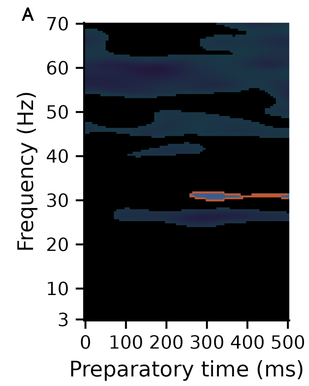
A. Broadband spontaneous preparatory activity decreased with age in both AS and VGS ($p<0.001$), with a significant main effect of condition ($p=0.02$) and an age-dependent modulation of spontaneous activity during inhibitory control trials ($p=0.02$). Decreases in spontaneous power were linked to **C.** reduced latency variability ($p=0.03$), but not to **B.** AS performance nor **D.** latency.

2. Mid-γ and high-β ERSP responses associated with improved AS performance and reduced variability



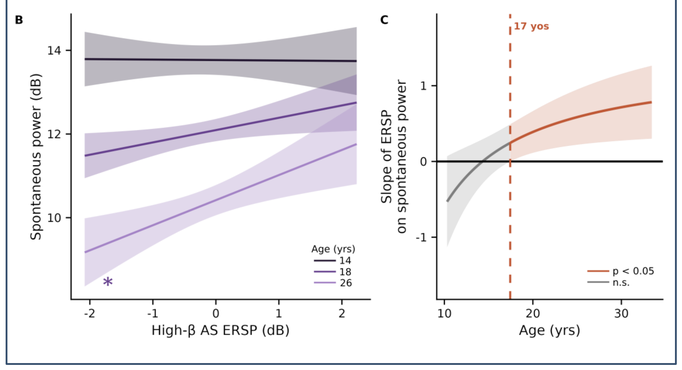
A. 5 ERSP clusters showed an age by condition interaction during the preparatory period, reflecting age-related change in the cognitive control aspects of the AS task. Two clusters were linked to AS performance: **B-C.** mid-γ (44-50 Hz) and **F-G.** high-β (30-32 Hz); both showed greater suppression on AS vs. VGS trials, with this difference decreasing developmentally. For mid-γ, reduced AS suppression was associated with **D.** improved AS performance ($p=0.003$) and **E.** lower variability ($p=0.002$). For high-β, **I. (left)** greater AS suppression was related to reduced variability ($p=0.004$), while **(right)** greater VGS suppression was associated with reduced variability, but only at younger ages ($p=0.02$).

3. Developmental increases in the strength of modulation between high-β ERSP and spontaneous power



A. The AS high-β ERSP cluster was related to spontaneous power, exhibiting a **B.** significant interaction with age ($p=0.03$); greater high-β suppression reduced spontaneous activity to an increasing extent across development.

C. Johnson-Neyman analysis revealed that this interaction was significant from age 17.4 and older. We found no relationship between mid-γ ERSP and spontaneous power.



CONCLUSIONS

- While spontaneous activity is important for maintaining flexible neural circuitry, during task, spontaneous activity may represent a "noise floor" of neural activity where high levels impede response inhibition by priming individuals to make a response[11], thus age-related decreases in spontaneous power during task may allow for higher fidelity transmission of task relevant signal allowing for developmental improvements in AS performance and reductions in variability.
- In comparison to adults, adolescents require greater task-related suppression of spontaneous power during AS (relative to VGS), which may suggest that adults have low enough spontaneous activity to transmit task relevant signal, whereas adolescents must suppress to a greater extent to perform the task correctly.
- Age-related increases in mid-γ activity may signal core aspects of task performance, but reflect aspects of task-evoked activity, rather than ongoing, spontaneous activity.
- Adolescents showed increased modulation of high-β activity during AS (relative to baseline VGS), suggesting neural circuits may be less readily primed for cognitive tasks and require greater high-β modulation when preparing an inhibitory response.
- Development of high-β activity may reflect maturational changes in prefrontal circuitry which serve to reduce levels of spontaneous activity[12], allowing for developmental improvements in both inhibitory control performance and reliability.
- **Together, these results illustrate a role for oscillatory prefrontal dynamics in supporting the stabilization of neural processing from adolescence into adulthood.**

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